

ANANTHA KRISHNA CHILAPPAGARI

GENERATIVE AI & AGENTIC SYSTEMS ENGINEER

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SUMMARY

- **AI/GenAI Engineer with 4+ years of experience** building production-ready backend, cloud, and agentic AI systems across AWS, enterprise APIs, document intelligence, and cybersecurity workflows.
- Built **LLM agents, RAG workflows, and MCP-based tool-calling systems** that convert natural-language requests into bounded, auditable actions over enterprise data and operational services.
- Strong hands-on depth in **Amazon Bedrock Agents, AWS Lambda, Step Functions, SageMaker, Textract, DynamoDB, FastAPI, Terraform**, and secure API integration.
- Experienced in designing **guardrailed execution layers** with schema validation, RBAC/JWT authentication, row limits, approval workflows, structured logging, CloudWatch monitoring, and human escalation paths.
- Proven ability to translate ambiguous business problems into **reliable AI products**, evaluation harnesses, synthetic test datasets, CI/CD checks, and measurable production outcomes.

EXPERIENCE

Gen AI / Backend Engineer | **SunPlusData**

Jan 2025 – Present

Atlanta, Georgia

- Designed and shipped an **AI assistant for a Physical Identity & Access Management platform**, using **MCP** to expose existing REST APIs and stored procedures as LLM-callable tools for natural-language queries over access records.
- Deployed the assistant as a **containerized FastAPI service** secured with **JWT authentication**, delivering sub-second, structured answers for security administrators.
- Built an **event-driven document-processing pipeline** on AWS where S3 uploads trigger CloudWatch/EventBridge events, Lambda document classification, and Step Functions extraction workflows.
- Routed each document to schema templates for structured extraction; when no template exists, invoked a **Bedrock agent** to generate a dynamic schema for previously unseen layouts.
- Used **Amazon Textract** for layout-aware OCR and table extraction, **Bedrock agents** to normalize outputs into validated fields, and **Amazon Rekognition** to detect embedded logos, stamps, and signatures.
- Implemented **RBAC**, multi-level approval workflows, status tracking, template mappings, and audit-ready logs before SAP submission for full traceability.
- Developed an **automated Pytest suite** to validate RBAC logic, approval workflows, and API endpoint behavior, strengthening reliability before production deployment.
- Developed a **data quality validation framework** with automated checks for schema compliance, completeness, and accuracy, ensuring data integrity before downstream processing.

Software Engineer | **Constella Intelligence Pvt. Ltd.**

Jan 2021 – July 2023

Hyderabad, India

- Designed and automated **event-driven ETL orchestration** on AWS Step Functions with CloudWatch monitoring, cutting end-to-end processing time by **40%**.
- Built **containerized ingestion services** on AWS ECS with Application Load Balancer autoscaling for high-volume file intake, reducing idle resource utilization by **20%**.
- Optimized large-scale data processing with **PySpark on Amazon EMR**, lowering operational costs by **15%**; automated cluster alerting contributed to **20%** overall cloud-spend reduction.
- Engineered an **ML-driven cybersecurity pipeline** that detects malicious emails with **93% accuracy**, using NLP feature engineering over email content and metadata.
- Developed a **Flask REST API** for real-time email risk scoring with async request handling, input validation, and structured logging for production observability.
- Re-platformed the malicious-email model to **Amazon SageMaker** using an XGBoost classifier, managed training jobs, hyperparameter tuning, versioned artifacts, and SageMaker Model Monitor for drift tracking.
- Built **CI/CD pipelines for ML workflows** in AWS Step Functions and integrated **AWS Lambda triggers** for real-time monitoring and alerting of build failures, improving deployment reliability.

PROJECTS

Suraksha – Agentic IT Support & Self-Healing Automation

Publication

AIS Hackathon Runner-Up · Amazon Bedrock (Claude 3.7 Sonnet), MCP, AWS Lambda, CloudWatch, Step Functions, DynamoDB

- Built an **agentic IT-support system** where Claude on Amazon Bedrock interprets natural-language requests and executes service-desk operations through **MCP tools**; project was an **AIS Hackathon runner-up** and publication-backed.
- Implemented a **self-healing loop** where monitoring agents track CloudWatch logs and metrics, detect anomalies, apply validated fixes from a knowledge base, and escalate novel incidents.
- Automated password resets and access updates using **AWS Lambda** and **Step Functions** under JWT authentication and RBAC, with a federated design that keeps sensitive logs local.

Conversational Data-Analyst Agent (Text-to-SQL)

Amazon Bedrock Agents, Amazon Athena, AWS Glue Data Catalog, Lambda, Terraform

- Built a **conversational analytics agent** that converts natural-language business questions into **validated SQL**, executes them against an **Amazon Athena** data lake, and returns explained results.
- Enforced a **guardrailed execution layer** with schema-aware query validation, read-only permissions, row limits, and a dry-run cost check so the agent’s actions stay bounded and safe.

Agent-Driven Adaptive Pregnancy Quiz & Flashcard Platform

Amazon Bedrock Knowledge Bases, Amazon OpenSearch, FastAPI, AWS EKS, DynamoDB, JWT, Docker

- Built a **personalized pregnancy-learning agent** that generates week-specific quizzes using **Amazon Bedrock Knowledge Bases** and retrieval-augmented generation over curated medical content, ensuring non-repetitive and context-aware question delivery.
- Developed an adaptive flashcard service that generates **three daily learning cards** based on user progress, quiz history, and spaced-repetition signals, storing interaction metadata in **Amazon DynamoDB** for personalized recommendations.
- Exposed quiz and flashcard workflows as secure **FastAPI REST microservices** with **JWT authentication** for user-profile management, protecting personalized learning endpoints and session-specific content generation.
- Containerized the services with **Docker** and deployed them on **AWS EKS**, enabling Kubernetes-based scaling, environment consistency, and reliable hosting for production-ready AI learning workflows.

SKILLS

Programming Languages	Python, SQL, Bash
Generative AI & Agents	Amazon Bedrock Agents, Knowledge Bases, Guardrails, Model Context Protocol (MCP), LangChain, Lang-Graph, RAG, prompt engineering, tool calling
AWS AI/ML	SageMaker, Textract, Rekognition, Comprehend Medical, OpenSearch
AWS Cloud & Data	Lambda, Step Functions, ECS, EKS, EMR, S3, DynamoDB, CloudWatch, API Gateway
Backend & APIs	FastAPI, Flask, REST API design, microservices
Data Science & ML	NLP, feature engineering, fraud detection, anomaly detection, predictive modeling, model monitoring, drift detection
DevOps & Security	Docker, Kubernetes, GitHub Actions, CI/CD, Pytest, JWT, OAuth 2.0, RBAC
Evaluation	Query correctness scoring, answer faithfulness, retrieval precision, BLEU, ROUGE, F1, regression test suites, LLM as Judge

EDUCATION

University of North Texas – M.S. in Data Science	CGPA: 3.99/4.0
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